

Special right triangles



1 In a $30^\circ - 60^\circ - 90^\circ$ triangle, identify which leg is shorter.

2 Find the exact value of the following:

a $\sin 60^\circ$

b $\cos 60^\circ$

c $\tan 30^\circ$

d $\sin 45^\circ$

3 Consider the following equilateral triangle.
Find the exact values of the following:

a Perpendicular height of the triangle

b Size of angle x

c

i $\sin 60^\circ$

ii $\cos 60^\circ$

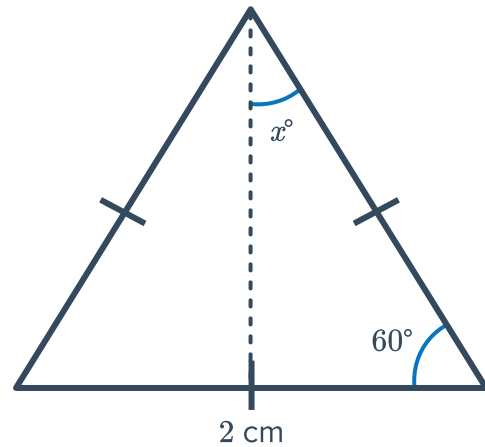
iii $\tan 60^\circ$

d

i $\sin x$

ii $\cos x$

iii $\tan x$



Exact value trigonometric ratios

4 Consider $\sin \theta = \frac{\sqrt{3}}{2}$.

a Find the value of θ , given that $0^\circ \leq \theta \leq 90^\circ$.

b Now, find the value of $\cos \theta$.

5 Consider $\sin \theta = \frac{1}{\sqrt{2}}$.

a Find the value of θ , given that $0^\circ \leq \theta \leq 90^\circ$.

b Now, find the exact value of $\tan \theta$.

6 Consider $\cos \theta = \frac{\sqrt{3}}{2}$.



a Find the value of θ , given that $0^\circ \leq \theta \leq 90^\circ$.

b Now, find the value of $\sin \theta$.

7 Consider $\cos \theta = \frac{1}{2}$.

a Find the value of θ , given that $0^\circ \leq \theta \leq 90^\circ$.

b Now, find the value of $\tan \theta$, correct to one decimal place.

8 Consider $\tan \theta = 1$.

a Find the value of θ , given that $0^\circ \leq \theta \leq 90^\circ$.

b Now, find the value of $\sin \theta$, correct to two decimal places.

9 Consider $\tan \theta = \frac{1}{\sqrt{3}}$.

a Find the value of θ , given that $0^\circ \leq \theta \leq 90^\circ$.

b Now, find the value of $\sin \theta$.

10 Consider $\tan \theta = \frac{1}{\sqrt{3}}$.

a Find the value of θ , given that $0^\circ \leq \theta \leq 90^\circ$.

b Now, find the value of $\cos \theta$, correct to one decimal place.

11 Consider $\cos \theta = \frac{\sqrt{3}}{2}$.

a Find the value of θ , given that $0^\circ \leq \theta \leq 90^\circ$.

b Now, find the value of $\tan \theta$, correct to two decimal places.

12 Consider $\sin \theta = \frac{\sqrt{3}}{2}$.

a Find the value of θ , given that $0^\circ \leq \theta \leq 90^\circ$.

b Now, find the value of $\tan \theta$, correct to two decimal places.

13 Consider $\tan \theta = \sqrt{3}$.



- a Find the value of θ , given that $0^\circ \leq \theta \leq 90^\circ$.
- b Now, find the value of $\sin 60^\circ$, correct to one decimal place.

14 For each of the following, find the value of θ given that $0^\circ \leq \theta \leq 90^\circ$:

a $\cos \theta = \frac{1}{\sqrt{2}}$

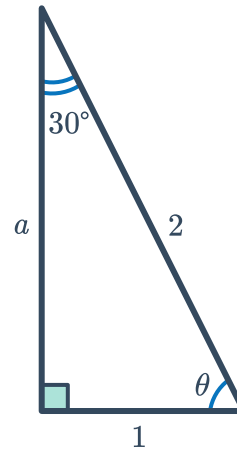
b $\tan \theta = \sqrt{3}$

c $\sin \theta = \frac{\sqrt{3}}{2}$

d $\tan \theta = \frac{1}{\sqrt{3}}$

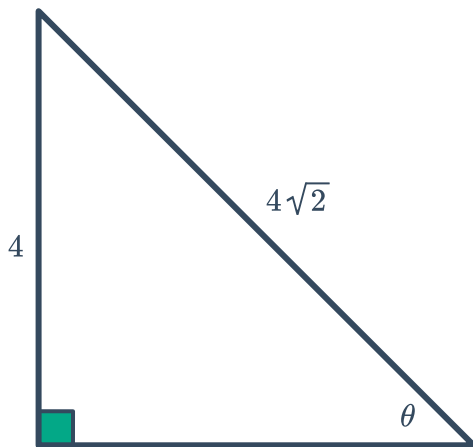
15 Consider the following triangle:

- a Find the exact value of a .
- b Find the value of θ .

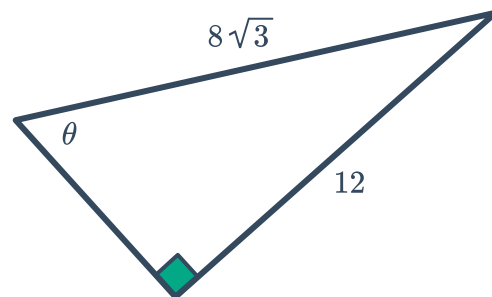


16 Use exact values to solve for θ in the following triangles:

a



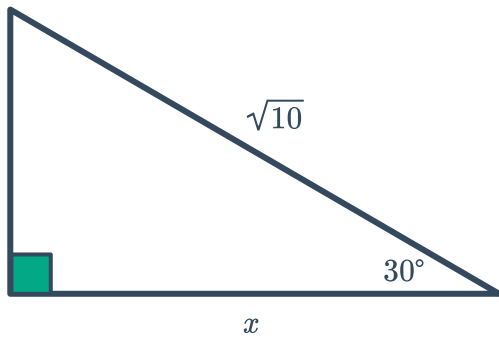
b



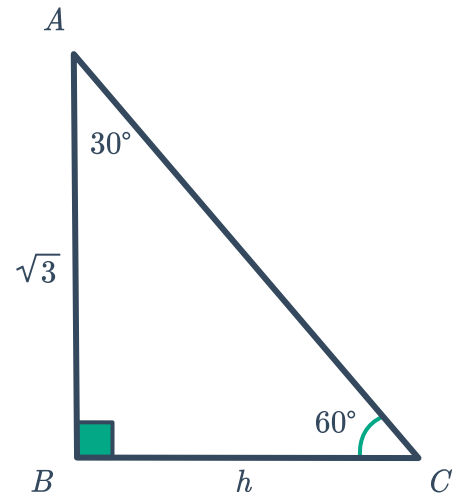
17 Find the exact value of the variable(s) in the following triangles:



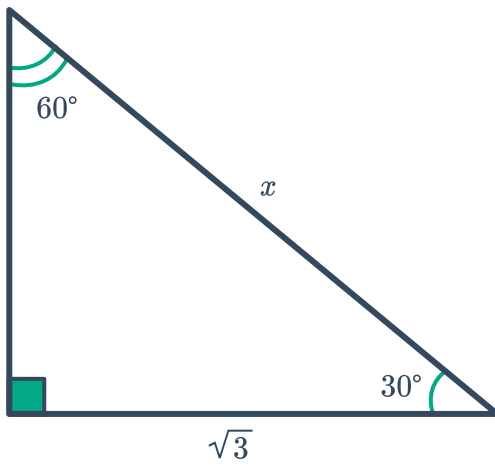
a



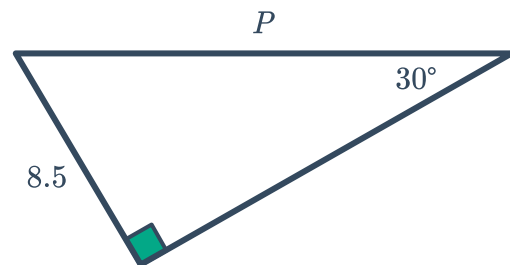
b



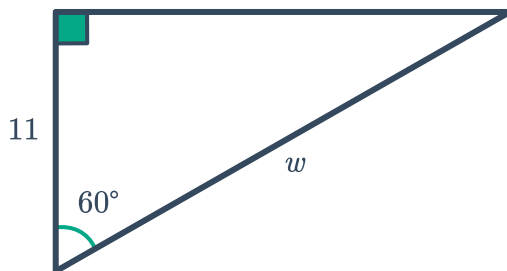
c



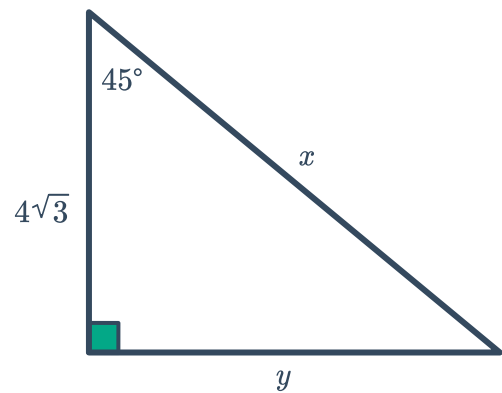
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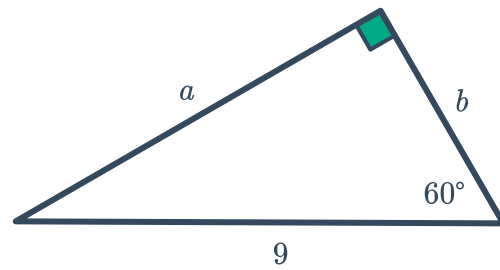
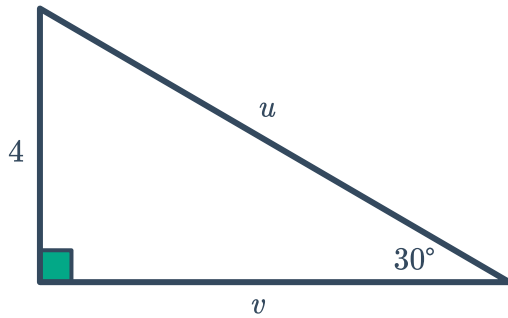
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f

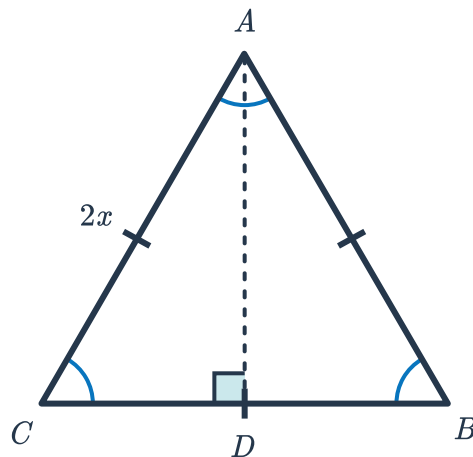


g



Applications

- 18** $\triangle ABC$ is an equilateral triangle with side lengths of $2x$ units. An altitude is drawn from A to meet side $\overline{BC}$ at D . This information is shown in the diagram below.



- Find the $m\angle B$.
- Given that $\overline{AD}$ bisects $\angle BAC$, find the measure of $\angle BAD$.
- Given that $\overline{AD}$ bisects $\overline{BC}$, find the length of $\overline{BD}$.
- Determine the length of the altitude $\overline{AD}$ in terms of x .
- State whether the following statements are true or false. If false, correct the statement so it is true.
 - The longer leg of a 30° - 60° - 90° triangle is twice the length of the shorter leg.
 - The shorter leg of a 30° - 60° - 90° triangle is opposite the angle measuring 30° .
 - The hypotenuse of a 30° - 60° - 90° triangle is $\sqrt{3}$ times the length of the longer leg.
 - The ratio of sides in a 30° - 60° - 90° triangle is $1 : \sqrt{3} : 2$.



19 Consider the figure shown:

a Find the exact length of the following:

i p ii q iii r iv s

b Now, find the exact perimeter of the quadrilateral.

